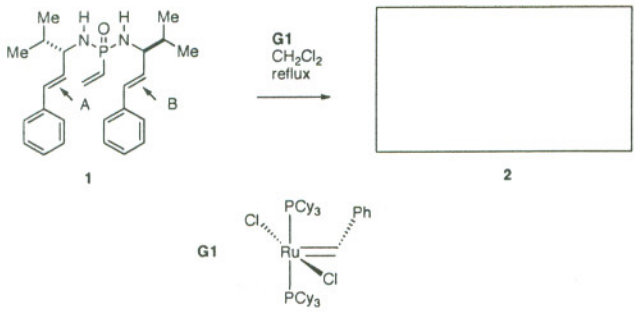
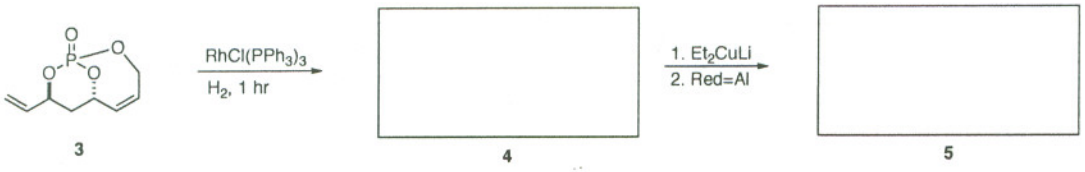
Organic Cume
10-2-09

1. Draw the structure of the product (**2**) in eq 1 (*Org. Lett.* **2000**, *2*, 1769-1772). Is the starting material (**1**) chiral? How about the product (**2**)? Are the two internal olefins (A and B) in **1** enantiotopic or diastereotopic?



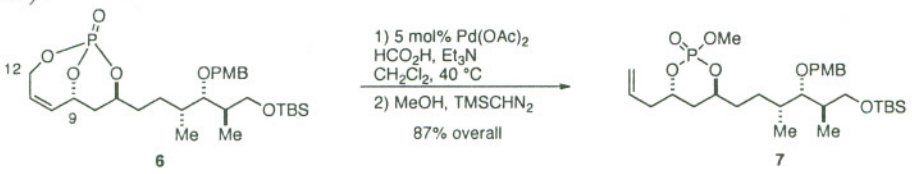
(1)

2. Provide the structures of compounds **4** and **5** (*Org. Lett.* **2005**, *7*, 3375-3378). Draw a clear, unambiguous three dimensional representation of compounds **3** and **4**. Use the three dimensional drawing of **4** to rationalize the stereochemistry of the cuprate addition (Et_2CuLi).



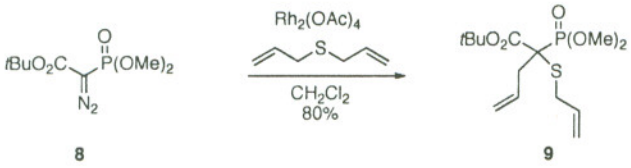
(2)

3. Write a mechanism for the transformation of **6** to **7** in eq 3. Draw a clear, unambiguous three dimensional representation of compound **6** and use it to rationalize the regioselectivity for the cleavage of the C-O bond at C12 over C9 (*Org. Lett.* **2008**, *10*, 109-112).



(3)

4. Write a mechanism for the transformation of **8** to **9** in eq 4. (*J. Org. Chem.* **2002**, *67*, 8123-8129).



(4)